

Roll No.

TEC-501

B. TECH. (ECE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
DIGITAL SIGNAL PROCESSING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Explain the basic elements of Digital Signal Processing System. Point the advantage of digital over analog signal processing. 10 Marks (CO1)

OR

- (b) Define Discrete Time (DT) sequence in general. Explain Even and Odd sequences. Determine the even and odd component of the following sequence : 10 Marks (CO1)

$$x(n) = \{-2, 1, \underset{\uparrow}{2}, -1, 3\}$$

2. (a) Illustrate the discrete time sinusoid properties with proof.

10 Marks (CO1)

OR

- (b) (i) Define 'Up-sampler' and prove that it is 'Time Varying'.
(ii) Determine the energy and power of the sequence :

$$x(n) = u(n)$$

10 Marks (CO1)

P. T. O.

3. (a) Determine the unit sample response i. e. $h(n)$ of the following, using Z-transform and its properties : 10 Marks (CO1)

(i) $y(n+2) + 3y(n+1) + 2y(n) = x(n+1) - x(n)$

(ii) $H(z) = \frac{z^2}{(z-1)^2} \cdot \frac{1}{(z+1)}$ for Causal system.

OR

- (b) (i) State and prove the "Parseval's Relation" for discrete time periodic sequence.

- (ii) Determine DTFT and Energy Density Spectrum (ESD) of a sequence as : 10 Marks (CO1)

$$x(n) = \begin{cases} A, & 0 < n < L-1 \\ 0, & \text{otherwise} \end{cases}$$

4. (a) (i) Prove the DFT as linear transformation. 10 Marks (CO2)

- (ii) Compute the 4-point DFT of given sequence :

$$x(n) = \{ \underset{\uparrow}{1}, 2, 1, 2 \}$$

OR

- (b) Determine the "Circular Convolution" of the following two sequences :

10 Marks (CO2)

$$x_1(n) = \{ \underset{\uparrow}{1}, 2, 3, 4 \} \text{ and } x_2(n) = \{ \underset{\uparrow}{2}, 1, 2, 1 \}$$

(3)

5. (a) Obtain and prove the 8-point butterfly flow graph for DITFFT (Decimation-in-time-Fast Fourier Transform). 10 Marks (CO2)

OR

- (b) Compute 8-point $X(k)$ of given below sequence by using DITFFT algorithm : 10 Marks (CO2)

$$x(n) = \{1, 2, 3, 4, 4, 3, 2, 1\}$$

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